



Shinrai
TECHNOLOGIES

YOUR TRUSTED CLOUD PARTNER

EKYC ENDPOINT CONSOLIDATION PROPOSAL

Electronic Know Your Customer Backend Engine

Jubilee
INSURANCE

SEPTEMBER 18, 2025

SHINRAI TECHNOLOGIES LIMITED

Ushuru Pension Plaza, Muthangari Drive, Westlands Nairobi

TECHNICAL STATEMENT OF WORK

Executive Summary

This technical specification outlines the consolidation of Jubilee's eKYC system from 10+ specialized endpoints into a single unified /kyc endpoint. The project prioritizes core functionality delivery and critical security fixes while implementing essential risk mitigation strategies.

Current State Analysis

Existing Endpoints (To Be Consolidated)

- Document Operations: /document/nationalid, /document/passport, /document/krapincertificate, /document/cr12
- Government Verification: /government/nationalid, /government/passport
- Other Operations: /background-check, /face-liveness, /agent-registration, /customer-registration
- Supporting: /stream/{bucketType}/{documentKey+}

Critical Security Issues (Must Address)

1. Lack of 2FA - Account takeover risk
2. Missing Security Headers - XSS vulnerability
3. Insecure Refresh Token Handling - Missing HttpOnly flags

Technical Scope & Implementation

Phase 1: Security-First Architecture (Days 1-3)

1.1 Critical Security Implementation (Day 1)

Technical Deliverables:

- Security headers configuration
- HttpOnly cookie setup for refresh tokens
- 2FA integration with existing Cognito

Technical Activities:

- Implement essential security headers: Strict-Transport-Security, Content-Security-Policy, X-Frame-Options
- Configure HttpOnly flags for Cognito refresh tokens
- Integrate 2FA with current authentication flow
- PRIORITY: Security fixes deployed to staging by end of Day 1

1.2 Unified API Design (Days 2-3)

Technical Deliverables:

- Single /kyc endpoint specification
- Action-based routing schema
- Unified request/response format
- Basic feature flag framework

Technical Activities:

- Design action parameter structure (action: "validate_nationalid", etc.)
- Create standardized payload format for all 10+ operations
- Define consistent response structure
- Implement simple feature flags for operation enable/disable

- Document core API specification

Phase 2: Backend Implementation (Days 4-7)

2.1 Core Consolidation (Days 4-5)

Technical Deliverables:

- Single Lambda function with action routing
- Consolidated service handlers
- Basic error handling

Technical Activities:

- Develop action-based request router
- Refactor existing handlers into modular functions
- Implement switch logic for all operation types
- Create unified response formatter
- Basic logging and error handling

2.2 Service Integration (Days 6-7)

Technical Deliverables:

- All 10+ operations integrated
- Legacy proxy layer
- Performance optimization

Technical Activities:

- Integrate all document validation services
- Consolidate government verification endpoints
- Merge background check and face liveness
- Implement registration workflows
- Create legacy endpoint proxies for backward compatibility
- Basic performance optimization

Phase 3: Deployment & Testing (Days 8-10)

3.1 API Gateway & Testing (Days 8-9)

Technical Deliverables:

- Configured API Gateway
- Core test suite
- Staging deployment

Technical Activities:

- Configure single /kyc endpoint in API Gateway
- Implement request validation and CORS
- Create essential unit and integration tests
- Focus testing on critical paths only
- Deploy to staging with health checks

3.2 Production Deployment (Day 10)

Technical Deliverables:

- Production deployment
- Basic monitoring
- Essential documentation

Technical Activities:

- Production deployment with feature flags
- Configure basic monitoring (CloudWatch)
- Create essential API documentation
- Implement rollback procedures
- Project handover and closure

Legacy Proxy Implementation

- All existing endpoints remain functional
- Requests automatically routed to new unified endpoint
- Maintains backward compatibility during client migration
- Gradual deprecation timeline post-delivery

Technical Deliverables**Core Technical Components**

1. Single /kyc Endpoint - Handles all 10+ operations via action parameter
2. Security Implementation - All 3 critical vulnerabilities resolved
3. Unified Lambda Function - Action-based routing with error handling
4. Legacy Proxy Layer - Backward compatibility during transition
5. Basic Feature Flags - Operation enable/disable capability
6. Essential Test Suite - Core functionality validation
7. Production Deployment - With basic monitoring and rollback capability

Documentation Deliverables

1. API Reference - Core endpoint documentation with examples
2. Migration Guide - Step-by-step client transition instructions
3. Security Report - Documentation of implemented security fixes
4. Rollback Procedures - Emergency rollback instructions

Excluded from Technical Scope

- Comprehensive monitoring dashboards
- Advanced performance optimization
- Extensive load testing
- Client SDKs
- Advanced feature flag management
- Detailed operational runbooks

Technical Success Criteria**Must-Have Technical Requirements**

- Single /kyc endpoint handles all 10+ operations
- All 3 security vulnerabilities resolved
- Response time <3 seconds average
- Legacy endpoints functional via proxy
- Basic rollback capability operational
- Error rate <2% during initial deployment

Should-Have Technical Requirements

- Feature flags for operation control
- Basic monitoring and alerting

- Essential API documentation
- Clear migration path for clients

Technical Risk Assessment

High-Priority Technical Risks

1. Integration Complexity

- **Risk: 10+ service integrations in compressed timeline**
- **Technical Mitigation:**
 - Reuse existing service handlers
 - Wrapper approach for legacy code
 - Parallel integration development

2. Single Endpoint Failure

- **Risk: All operations dependent on one endpoint**
- **Technical Mitigation:**
 - Legacy proxy layer for immediate fallback
 - Feature flags for operation-level control
 - Circuit breaker patterns for external services
 - CloudWatch monitoring with automated alerts

3. Security Implementation

- Risk: Security fixes affecting existing functionality
- Technical Mitigation:
 - Security changes implemented first (Day 1)
 - Isolated security configuration
 - Comprehensive security testing

Technical Assumptions

Infrastructure Assumptions

- AWS Lambda and API Gateway infrastructure available
- Existing Cognito authentication system functional
- All third-party service integrations (LexisNexis, IPRS) operational
- Development and staging environments accessible

Code Base Assumptions

- Current endpoint implementations are stable
- Existing service handlers can be refactored
- Database schemas remain unchanged
- No breaking changes to external service APIs

Technical Timeline

Week 1: Security + Core Development

Day	Technical Focus	Key Technical Deliverables
Day 1	Security Implementation	Security headers, HttpOnly cookies, 2FA integration
Day 2	API Design	Unified endpoint spec, action routing design
Day 3	Architecture Setup	Request/response formats, basic feature flags
Day 4	Core Backend	Lambda function, action router, error handling
Day 5	Service Integration	Document validation, government verification

Week 2: Integration + Deployment

Day	Technical Focus	Key Technical Deliverables
Day 6	Complete Integration	Background check, face liveness, registrations
Day 7	Legacy Compatibility	Proxy layer, backward compatibility testing
Day 8	API Gateway + Testing	Gateway config, essential test suite
Day 9	Staging Deployment	Full staging environment, integration testing
Day 10	Production Deployment	Live deployment, monitoring, handover

FINANCIAL PROPOSAL

Project Investment

Total Project Value: \$3,000 USD (exclusive of VAT)

Resource Allocation: 2 Senior Full-Stack Developer (10 days)

Budget Breakdown

Phase	Duration	Investment	Percentage	Deliverables
Advance Payment	-	\$600	20%	Resource Mobilization
Phase 1: Security & Architecture	3 days	\$720	24%	Security fixes + API design
Phase 2: Backend Implementation	4 days	\$960	32%	Core consolidation + integration
Phase 3: Deployment & Testing	3 days	\$720	24%	Essential testing + production
Total Investment	10 days	\$3,000	100%	Complete system consolidation

Investment Justification

Immediate ROI Benefits

1. Security Risk Mitigation - Eliminates critical vulnerabilities
2. Operational Efficiency - Single endpoint reduces integration complexity
3. Maintenance Cost Reduction - Consolidated codebase easier to maintain
4. Developer Experience - Simplified client integration process

Long-term Value Creation

1. Scalability - Single endpoint easier to scale and monitor
2. Future Enhancement - Consolidated architecture enables faster feature development
3. Compliance - Improved security posture supports regulatory requirements
4. Technical Debt Reduction - Eliminates maintenance of 10+ separate endpoints

Cost-Benefit Analysis

- Current State: 10+ endpoints requiring individual maintenance
- Future State: Single, secure, unified endpoint
- Maintenance Savings: Estimated 40% reduction in ongoing development costs
- Security Value: Risk mitigation worth significantly more than \$3,000 investment

Payment Terms

Payment Schedule

- **Advance Payment: \$600 USD (20%)**
 - Payable upon signed agreement
 - Enables resource mobilization and project setup
 - Covers initial project preparation
- **Phase 1 Completion: \$720 USD (24%)**
 - Payable upon security implementation completion
 - Covers critical security fixes and API design
 - Milestone-based payment for Phase 1 deliverables
- **Phase 2 Completion: \$960 USD (32%)**
 - Payable upon backend implementation completion
 - Covers core consolidation and service integration
 - Largest phase reflecting implementation complexity
- **Final Payment: \$720 USD (24%)**
 - Payable upon successful production deployment
 - Subject to technical acceptance criteria
 - Includes 7-day post-deployment support period

Payment Methods

- Bank transfer (preferred)

Currency & Tax

- All amounts in USD exclusive of VAT
- Client responsible for applicable taxes
- Currency conversion at prevailing rates

Contract Terms

Project Scope Boundaries

- Included: All items listed in Technical Statement of Work
- Excluded: Items explicitly marked as "Out of Scope"

- Changes: Additional scope requires separate agreement

Intellectual Property

- Client retains all rights to business logic and data
- Standard development tools and frameworks remain open source
- Custom integration code becomes client property upon payment

Support & Warranty

- Included: 7 days post-deployment support for bugs/issues
- Response Time: 24 hours for critical production issues
- Scope: Issues directly related to delivered functionality

Risk & Mitigation Investment

Included Risk Mitigation (No Additional Cost)

- Legacy proxy layer for zero-downtime transition
- Basic feature flags for operational control
- Essential rollback procedures
- Security vulnerability resolution

Risk-Adjusted Pricing

- Fixed price model absorbs development risks
- No additional charges for reasonable scope adjustments
- Performance guarantee included in base price

Value Proposition Summary

Investment: \$3,000 USD

Delivery Timeline: 10 business days

Value Delivered:

- Consolidated, secure eKYC system
- Eliminated security vulnerabilities
- Simplified client integration
- Reduced maintenance overhead
- Zero-downtime migration capability

ROI Realization: Immediate security benefits, long-term operational savings

Acceptance & Next Steps

Project Approval Process

1. Technical Review - Stakeholder review of technical specifications
2. Financial Approval - Budget and payment terms confirmation
3. Contract Execution - Signed agreement and initial payment
4. Project Kickoff - Day 1 commencement with security implementation

Immediate Requirements

- Technical scope approval
- Financial terms acceptance
- AWS infrastructure access confirmation
- Initial payment processing
- Project stakeholder designation

REVIEW & SIGN OFF

NAME	ROLE	SIGNATURE	DATE
George Makori	Head of Digital Factory	Signed by: <i>George Makori</i> AC0F4CFA92824AE	19/9/2025
Alex Chege	Head of PMO & BPM	Signed by: <i>Alex Chege</i> 9A6BAEE00005404...	22/9/2025
Kigunda Kananua	Head of Operations- Retail & Group Life	Signed by: <i>Kigunda Kananua</i> AA9F05D94541490...	19/9/2025
Anna Manyara	COO	Signed by: <i>Anna Manyara</i> CAE43345F78D4F7	22/9/2025

APPROVAL

NAME	ROLE	SIGNATURE	DATE
Asman Mugambi	CEO	Signed by: <i>Asman Mugambi</i> 6380D6097432412	19/9/2025
Timothy Munyao	CEO & Founder – Shinrai Technologies	Signed by: <i>Timothy Munyao</i> B9C4CABD687348A...	19/9/2025